

A Matched Seven-Likelihood Test of a Zero-Particle-CDM Single-Metric Relativistic Modified-Gravity Cosmology

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We test a single-metric relativistic modified-gravity cosmology in which the particle cold-dark-matter density and the bare cosmological constant are fixed to zero. The theory combines an aether scalar–tensor sector, a current-linked nonseparable scalar interaction, and an auxiliary Type-II minimally modified gravity sector for late-time acceleration. The realization tested here is the cyclic, nonlinear realization of this framework rather than the earlier linearized realization. The frozen implementation is compared with a matched flat- Λ CDM control using the same Planck 2018 high- ℓ TTTEEE-lite, low- ℓ TT, low- ℓ EE, Planck lensing, DESI DR2 BAO, Pantheon+, and SH0ES Gaussian likelihood terms. Both Markov chains satisfy the common convergence requirement, with $R - 1 = 0.00691$ for modified gravity and 0.00729 for Λ CDM. The lowest sampled independent seven-likelihood totals are $\chi^2_{\text{MG}} = 2441.26650$ and $\chi^2_{\Lambda\text{CDM}} = 2446.93250$, giving $\Delta\chi^2 = -5.66600$. With seven fitted cosmological coordinates for modified gravity and six for Λ CDM, $\Delta\text{AIC} = -3.66600$ and, for the stated bookkeeping convention $N = 2393$, $\Delta\text{BIC} = +2.11431$. The modified-gravity posterior gives $H_0 = 68.672 \pm 0.305 \text{ km s}^{-1} \text{ Mpc}^{-1}$, compared with 68.835 ± 0.273 for Λ CDM. The likelihood improvement is concentrated in Planck high- ℓ , Pantheon+, and DESI DR2, while the SH0ES contribution changes by $+1.452$ in $\Delta\chi^2$. These results establish that the tested zero-particle-CDM realization is statistically competitive with the matched Λ CDM control on this likelihood stack. Complete analytic and numerical derivations used by the implementation are collected in the appendices.

I. INTRODUCTION

The standard Λ CDM cosmology remains the benchmark description of the cosmic microwave background (CMB), baryon acoustic oscillations (BAO), large-scale structure, and the late-time expansion history [1–3]. The local distance-ladder determination of the Hubble constant continues to motivate tests of extended cosmological models [4, 5], while DESI DR2 has sharpened constraints on the expansion history and on phenomenological departures from a cosmological constant [3, 6–8].

Recent work has explored early dark energy, evolving dark energy, Type-II minimally modified gravity, sign-changing acceleration sectors, and relativistic modified gravity [9–16]. Model ranking depends on the data combination, parameterization, nuisance treatment, and parameter count, so matched comparisons remain the cleanest way to isolate the statistical performance of a particular construction.

The model considered here belongs to the Aether Scalar–Tensor (AeST) family [17, 18]. Its nonlinear Hamiltonian structure, cosmological dynamics, quasistatic solutions, cluster phenomenology, and compact objects have been investigated in a growing literature [19–24]. The specific realization tested here uses a nonseparable scalar interaction linked to the conserved scalar current and an auxiliary Type-II acceleration sector. Its parameters and field equations were fixed before the matched likelihood

comparison presented below.

The cosmological construction studied here is the cyclic, nonlinear realization of the framework. References below to linear scalar perturbations and linear matter spectra refer to the perturbative treatment used to compute observables and do not characterize the underlying model as a linear theory.

The zero-particle-CDM construction is also motivated by the tight empirical relation between baryonic distributions and galactic accelerations, first emphasized by MOND and quantified in modern galaxy samples [25–28]. The central question addressed here is whether the same relativistic gravitational framework can remain competitive on precision cosmological data when ω_{cdm} is fixed to zero.

II. RELATIVISTIC MODEL

Ordinary matter couples minimally and universally to a single metric $g_{\mu\nu}$. The gravitational sector contains a unit timelike vector A^μ and a shift-symmetric scalar ϕ . Define

$$F_{\mu\nu} = 2\nabla_{[\mu}A_{\nu]}, \quad J^\mu = A^\nu\nabla_\nu A^\mu, \quad (1)$$

$$q_{\mu\nu} = g_{\mu\nu} + A_\mu A_\nu, \quad \mathcal{Q} = A^\mu\nabla_\mu\phi, \quad (2)$$

$$\mathcal{Y} = q^{\mu\nu}\nabla_\mu\phi\nabla_\nu\phi. \quad (3)$$

The scalar–vector action is

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$$S_{\text{AeST}} = \frac{1}{16\pi\tilde{G}} \int d^4x \sqrt{-g} \left\{ R - \frac{K_B}{2} F_{\mu\nu} F^{\mu\nu} + (2 - K_B) (2J^\mu \nabla_\mu \phi - \mathcal{Y}) - \mathcal{F}(\mathcal{Y}, \mathcal{Q}) - \lambda(A^\mu A_\mu + 1) \right\} + S_m[g_{\mu\nu}, \psi_m]. \quad (4)$$

The scalar kinetic function is

$$\mathcal{F}(\mathcal{Y}, \mathcal{Q}) = \mathcal{F}_T(\mathcal{Q}) + \mathcal{Y}\Gamma(\mathcal{Q}) + J(\mathcal{Y}), \quad (5)$$

with

$$\mathcal{F}_T(\mathcal{Q}) = -4K_2 Z_0^2 \left[\cosh\left(\frac{\mathcal{Q} - \mathcal{Q}_0}{Z_0}\right) - 1 \right], \quad (6)$$

$$\Gamma(\mathcal{Q}) = \frac{2K_2 Z_0}{\mathcal{Q}} \sinh\left(\frac{\mathcal{Q} - \mathcal{Q}_0}{Z_0}\right). \quad (7)$$

The quasistatic function used in the screened galactic sector is

$$J(\mathcal{Y}) = \frac{a_0^2(2 - K_B)}{\beta_0^3} \left[-\frac{2\sqrt{\mathcal{Y}}\beta_0(\beta_0 + 1)}{a_0} + \frac{\mathcal{Y}\beta_0^2}{a_0^2} + 2(\beta_0 + 1)^2 \ln\left(1 + \frac{\sqrt{\mathcal{Y}}\beta_0}{a_0(\beta_0 + 1)}\right) \right]. \quad (8)$$

The mixed term is fixed by the homogeneous scalar current rather than fitted as an independent function. On FLRW, $\mathcal{Y} = 0$; at the screened static point, $\mathcal{Q} = \mathcal{Q}_0$ and $\Gamma(\mathcal{Q}_0) = 0$.

Late-time acceleration is supplied by an auxiliary Type-II minimally modified gravity sector of VCDM form [11–15]. Its relevant ADM contribution may be written

$$S_V = M_{\text{Pl}}^2 \int dt d^3x N \sqrt{\gamma} \left[-V(\varphi_V) - \frac{3}{4} \lambda_C^2 - \lambda_C(K + \varphi_V) - \mathcal{L}_{\text{gf}} \right]. \quad (9)$$

Eliminating the multiplier gives

$$\lambda_C = -\frac{2}{3}(K + \varphi_V), \quad (10)$$

and

$$\mathcal{L}_{V,\text{red}} = M_{\text{Pl}}^2 \left[\frac{(K + \varphi_V)^2}{3} - V(\varphi_V) - \mathcal{L}_{\text{gf}} \right]. \quad (11)$$

For the cosmological realization tested below,

$$\omega_{\text{cdm}} = 0, \quad \Omega_{\Lambda, \text{bare}} = 0, \quad (12)$$

while the effective cold cosmological component is carried by the scalar–vector sector. Detailed background, perturbative, constraint, and numerical derivations are given in Appendices A–J.

III. LIKELIHOODS AND MATCHED INFERENCE

A. Likelihood stack

The comparison uses seven independent contributions: Planck 2018 high- ℓ TTTEEE-lite, Planck low- ℓ TT, Planck low- ℓ EE, Planck lensing, DESI DR2 BAO [3], Pantheon+ supernovae [29, 30], and a Gaussian SH0ES term based on $H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [4]. The Boltzmann calculation uses CLASS [31, 32] and sampling uses Cobaya [33].

The SH0ES contribution is

$$\chi_{\text{SH0ES}}^2 = \left(\frac{H_0 - 73.04}{1.04} \right)^2. \quad (13)$$

The stored chain values at the minimum-total points are reproduced by Eq. (13): 16.0901 for the modified-gravity point $H_0 = 68.8683$ and 14.6382 for the Λ CDM point $H_0 = 69.0610$.

B. Parameter counts and convergence

The modified-gravity chain varies seven cosmological coordinates: H_0 , ω_b , ω_{aest} , one independent acceleration-history shape coefficient, n_s , A_s , and τ_{reio} . The matched flat- Λ CDM control varies the corresponding six standard cosmological coordinates, with ω_{cdm} in place of the AeST/acceleration pair. Planck calibration is treated identically in both runs.

The final convergence values are

$$R - 1 = 0.00691196 \quad (\text{modified gravity}), \quad (14)$$

$$R - 1 = 0.00728902 \quad (\Lambda\text{CDM}), \quad (15)$$

with a common stopping threshold $R - 1 < 0.01$. The comparison uses the seven independent likelihood columns stored in the converged chains.

The information criteria are

$$\text{AIC} = \chi^2 + 2k, \quad (16)$$

$$\text{BIC} = \chi^2 + k \ln N, \quad (17)$$

with $k_{\text{MG}} = 7$, $k_{\Lambda\text{CDM}} = 6$, and the stated bookkeeping convention $N = 2393$ [34–36].

IV. RESULTS

A. Matched likelihood comparison

The lowest sampled independent seven-likelihood totals are

$$\chi^2_{\text{MG}} = 2441.2664987, \quad (18)$$

$$\chi^2_{\Lambda\text{CDM}} = 2446.9324958, \quad (19)$$

so

$$\Delta\chi^2 \equiv \chi^2_{\text{MG}} - \chi^2_{\Lambda\text{CDM}} = -5.6659971. \quad (20)$$

The corresponding information-criterion differences are

$$\Delta\text{AIC} = -3.6659971, \quad (21)$$

$$\Delta\text{BIC} = +2.1143060. \quad (22)$$

The maximum sampled likelihood and AIC favor the modified-gravity realization, whereas the stated BIC convention gives a mild preference to the lower-dimensional ΛCDM control.

TABLE I. Independent likelihood contributions at the lowest sampled total in each converged chain. Negative $\Delta\chi^2_i$ favors modified gravity.

Likelihood	MG	ΛCDM	$\Delta\chi^2_i$
Planck high- ℓ	583.3074	586.8745	-3.5670
Planck lowTT	22.6653	22.7313	-0.0661
Planck lowEE	395.8339	396.1093	-0.2754
Planck lensing	8.7693	8.8027	-0.0334
DESI DR2	9.2113	10.4194	-1.2081
Pantheon+	1405.3892	1407.3572	-1.9680
SH0ES	16.0901	14.6382	+1.4519
Total	2441.2665	2446.9325	-5.6660

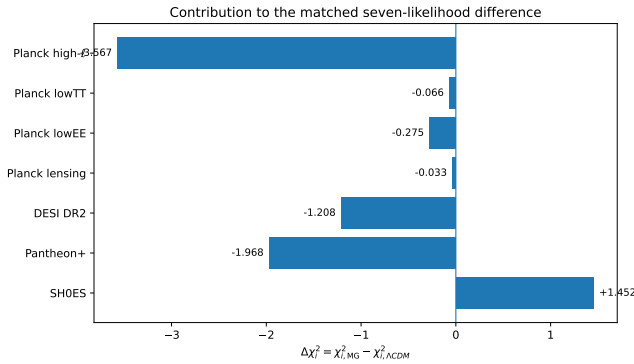


FIG. 1. Per-likelihood contribution to $\Delta\chi^2 = \chi^2_{\text{MG}} - \chi^2_{\Lambda\text{CDM}}$.

The largest contributions to the improvement are Planck high- ℓ (-3.567), Pantheon+ (-1.968), and DESI DR2 (-1.208). The SH0ES term contributes $+1.452$, so the net likelihood gain is generated by the CMB, BAO, and supernova sectors rather than by the local-distance-ladder term.

B. Posterior parameters

The marginalized Hubble constants are

$$H_0^{\text{MG}} = 68.6723 \pm 0.3050 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (23)$$

$$H_0^{\Lambda\text{CDM}} = 68.8349 \pm 0.2728 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (24)$$

The minimum- χ^2 sampled values are 68.8683 and 69.0610, respectively.

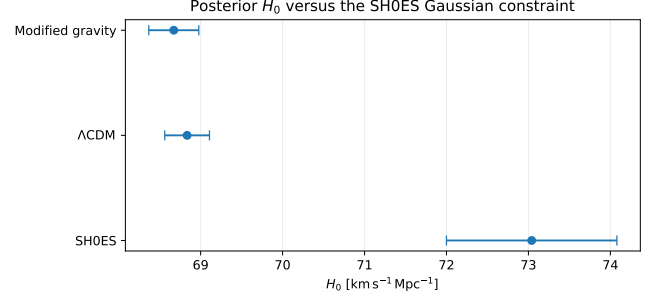


FIG. 2. Posterior H_0 values for the two matched cosmological chains compared with the Gaussian SH0ES determination used in the likelihood. Error bars show one standard deviation.

TABLE II. Selected posterior means and one-standard-deviation uncertainties.

Parameter	MG	ΛCDM
H_0	68.6723 ± 0.3050	68.8349 ± 0.2728
ω_b	0.0225242 ± 0.0001239	0.0225396 ± 0.0001201
ω_{cold}	0.118215 ± 0.000633	0.117916 ± 0.000627
n_s	0.969057 ± 0.003293	0.969428 ± 0.003347
$10^9 A_s$	2.09763 ± 0.02836	2.11265 ± 0.02827
τ_{reio}	0.05574 ± 0.00666	0.05929 ± 0.00661

Here ω_{cold} denotes ω_{aest} for modified gravity and ω_{cdm} for ΛCDM .

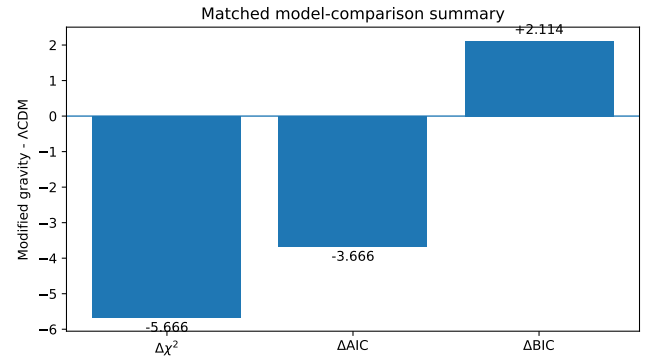


FIG. 3. Matched model-comparison summary. Negative values favor the modified-gravity realization; positive values favor ΛCDM .

C. Prior-boundary diagnostic

The sampling prior was $66 < H_0 < 70.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The posterior means lie well inside this interval. Relative to the marginalized standard deviations, the modified-gravity posterior mean is 8.76σ above the lower edge and 5.99σ below the upper edge; the corresponding ΛCDM distances are 10.39σ and 6.10σ .

A fixed-coordinate profile, in which H_0 alone was varied from 66 to 76 while the remaining coordinates were held at the lowest sampled point, rises sharply away from the minimum. At the upper sampling boundary $H_0 = 70.5$, the profile increases by $\Delta\chi^2 = 331.03$ for modified gravity and 258.26 for ΛCDM relative to the corresponding replay point. The profile minimum coincides with the chain-minimum H_0 for both models. These diagnostics show that the reported posterior region is well separated from the imposed H_0 boundaries.

V. CONTEXT AMONG RECENT COSMOLOGICAL EXTENSIONS

The present $\Delta\chi^2 = -5.666$ is a matched result for one specific seven-likelihood stack. Recent early-dark-energy analyses using different CMB combinations and larger parameter spaces have reported substantially larger raw likelihood improvements when SH0ES is included [9]. DESI DR2 analyses of the two-parameter $w_0 w_a$ extension likewise find a preference whose size depends on the supernova sample and statistical treatment [3, 6–8]. Other recent analyses examine phantom-crossing or sign-switch acceleration histories in related late-time data combinations [10, 16].

The distinguishing feature of the present comparison is structural rather than a record-sized likelihood shift: the model is evaluated with $\omega_{\text{cdm}} = 0$ and only one more fitted cosmological coordinate than the matched flat- ΛCDM control. On the stated likelihood stack, that construction attains a lower sampled χ^2 and AIC while retaining a BIC difference of order unity.

VI. CONCLUSIONS

A zero-particle-CDM single-metric relativistic modified-gravity cosmology has been tested against a matched flat- ΛCDM control using Planck 2018 high- ℓ and low- ℓ CMB likelihoods, Planck lensing, DESI DR2 BAO, Pantheon+, and SH0ES. Both chains satisfy the common convergence criterion $R - 1 < 0.01$. The lowest sampled totals give

$$\Delta\chi^2 = -5.6660, \quad (25)$$

with

$$\Delta\text{AIC} = -3.6660, \quad \Delta\text{BIC} = +2.1143 \quad (26)$$

under the stated parameter-count and $N = 2393$ conventions. The improvement is concentrated in Planck high- ℓ ,

Pantheon+, and DESI DR2. The inferred Hubble constants are 68.672 ± 0.305 and $68.835 \pm 0.273 \text{ km s}^{-1} \text{ Mpc}^{-1}$ for modified gravity and ΛCDM , respectively.

The matched likelihood comparison therefore establishes statistical competitiveness of the tested zero-particle-CDM realization on this data combination. The appendices provide the action, background reduction, perturbative mapping, stability analysis, constraint count, canonical normalization, numerical implementation, likelihood construction, prior diagnostic, and reproducibility ledger used to obtain the result.

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ARTIFICIAL-INTELLIGENCE ASSISTANCE DISCLOSURE

Substantive artificial-intelligence assistance was provided by OpenAI ChatGPT and Google Antigravity/Gemini tools for algebraic cross-checking, code generation and debugging, workflow organization, literature organization, and manuscript language editing. All numerical calculations reported in this article were executed in the author’s computing environment. The author reviewed the equations, numerical outputs, citations, and scientific claims and assumes full responsibility for the accuracy and integrity of the work.

DATA AND CODE AVAILABILITY

The modified CLASS implementation, exact run configurations, chain roots, independent likelihood extractor, convergence records, runtime SH0ES injection scripts, figure-generation code, and SHA-256 manifests are prepared for archival release. The development repository is <https://github.com/KarmirisP/aest-nonseparable-cosmology>.

Appendix A: Action variation and scalar derivatives

We use metric signature $(-, +, +, +)$. From Eq. (5),

$$F_{,\mathcal{Y}} = \Gamma(\mathcal{Q}) + J_{,\mathcal{Y}}, \quad (A1)$$

$$F_{,\mathcal{Y}\mathcal{Q}} = \Gamma_{,\mathcal{Q}}, \quad (A2)$$

$$F_{,\mathcal{Q}} = \mathcal{F}_{T,\mathcal{Q}} + \mathcal{Y}\Gamma_{,\mathcal{Q}}, \quad (A3)$$

$$F_{,\mathcal{Q}\mathcal{Q}} = \mathcal{F}_{T,\mathcal{Q}\mathcal{Q}} + \mathcal{Y}\Gamma_{,\mathcal{Q}\mathcal{Q}}. \quad (A4)$$

On FLRW, $\mathcal{Y} = 0$, while at the static screened point $\mathcal{Q} = \mathcal{Q}_0$ and $\Gamma(\mathcal{Q}_0) = 0$. The temporal derivatives required by

the background and perturbation systems are

$$\mathcal{F}_{T,\mathcal{Q}} = -4K_2 Z_0 \sinh\left(\frac{\mathcal{Q} - \mathcal{Q}_0}{Z_0}\right), \quad (\text{A5})$$

$$\mathcal{F}_{T,\mathcal{Q}\mathcal{Q}} = -4K_2 \cosh\left(\frac{\mathcal{Q} - \mathcal{Q}_0}{Z_0}\right), \quad (\text{A6})$$

$$\mathcal{F}_{T,\mathcal{Q}\mathcal{Q}\mathcal{Q}} = -\frac{4K_2}{Z_0} \sinh\left(\frac{\mathcal{Q} - \mathcal{Q}_0}{Z_0}\right), \quad (\text{A7})$$

and

$$\Gamma_{,\mathcal{Q}} = \frac{2K_2}{\mathcal{Q}^2} \left[\mathcal{Q} \cosh\left(\frac{\mathcal{Q} - \mathcal{Q}_0}{Z_0}\right) - Z_0 \sinh\left(\frac{\mathcal{Q} - \mathcal{Q}_0}{Z_0}\right) \right]. \quad (\text{A8})$$

Varying the Type-II auxiliary multiplier in Eq. (9) gives

$$-\frac{3}{2}\lambda_C - (K + \varphi_V) = 0, \quad (\text{A9})$$

which yields Eq. (11). The shift symmetry of ϕ gives a conserved scalar current $\nabla_\mu \mathcal{J}_\phi^\mu = 0$.

Appendix B: FLRW reduction and conserved current

For a spatially homogeneous scalar, $\mathcal{Y} = 0$. The shift-symmetric scalar equation integrates once to

$$I_0 = a^3 \mathcal{Q} F_{,\mathcal{Y}}. \quad (\text{B1})$$

With Eq. (7),

$$I_0 = 2K_2 Z_0 a^3 \sinh\left(\frac{\mathcal{Q} - \mathcal{Q}_0}{Z_0}\right). \quad (\text{B2})$$

Defining $x_0 = I_0/(2K_2 Z_0)$ gives the exact solution

$$\mathcal{Q}(a) = \mathcal{Q}_0 + Z_0 \operatorname{asinh}\left(\frac{x_0}{a^3}\right), \quad (\text{B3})$$

and $I_0 = 2K_2 Z_0 x_0$. The implementation uses $K_2 = 7500$, $\mathcal{Q}_0 = 0.1$, $Z_0 = 10^{-9}$, $x_0 = 0.02672752$, and $I_0 = 4.009128 \times 10^{-7}$ in solver units.

Since $\mathcal{Y} = 0$ exactly, the mixed operator does not alter $\mathcal{F}_T$, $\mathcal{F}_{T,\mathcal{Q}}$, or $\mathcal{F}_{T,\mathcal{Q}\mathcal{Q}}$ on FLRW. The identity

$$K_{\mathcal{Q}} = \mathcal{Q} F_{,\mathcal{Y}} = \frac{I_0}{a^3} \quad (\text{B4})$$

is imposed analytically in the numerical source.

Appendix C: Radiation-era scalar stability

The reduced radiation-era scalar characteristic polynomial is

$$A\lambda^2 + B\lambda + C = 0, \quad (\text{C1})$$

with

$$A = \frac{2I_0 K_B}{\mathcal{Q} a^3} - 4K_B + 8, \quad (\text{C2})$$

$$B = HA, \quad (\text{C3})$$

$$C = \frac{2HI_0(2 - K_B)}{a^3}. \quad (\text{C4})$$

On the expanding branch $H > 0$, $I_0 > 0$, $\mathcal{Q} > 0$, $a > 0$, and $0 < K_B < 2$, all three coefficients are positive. The high-frequency scalar sound speed is

$$c_s^2 = \frac{4K_2 K_B Z_0 \sinh[(\mathcal{Q} - \mathcal{Q}_0)/Z_0]/\mathcal{Q} + 4(2 - K_B)}{4K_2 K_B \cosh[(\mathcal{Q} - \mathcal{Q}_0)/Z_0]}. \quad (\text{C5})$$

The numerical branch used in the likelihood calculation was checked for positive c_s^2 , finite radiation-era histories, finite CMB spectra through $\ell = 2500$, and finite positive linear matter spectra over the regression range.

Appendix D: Linear scalar perturbations and CLASS mapping

In the perturbation variables used during the variation,

$$\theta_s = \frac{\varphi}{\mathcal{Q}}, \quad \alpha = \frac{\chi}{\mathcal{Q}} - \theta_s, \quad (\text{D1})$$

so $\varphi + \mathcal{Q}\alpha = \chi$. The quadratic contribution from the nonseparable operator becomes

$$\Delta\mathcal{L}^{(2)} = -\Gamma(\mathcal{Q}) \frac{k^2}{a^2} \chi^2. \quad (\text{D2})$$

The generic direct source entering the E equation contains

$$K_{\mathcal{Q}}\chi - \mathcal{Q}F_{,\mathcal{Y}}\chi - (2 - K_B)B. \quad (\text{D3})$$

Using Eq. (B4), the direct χ contribution cancels exactly, giving

$$K_B(\dot{E} + HE) = -(2 - K_B)B \quad (\text{D4})$$

for this source term. The mixed derivative $F_{,\mathcal{Y}\mathcal{Q}} = \Gamma_{,\mathcal{Q}}$ first contributes at cubic order. The auxiliary acceleration variables are eliminated algebraically at nonzero k , leaving the scalar–vector propagating basis after the metric constraints are solved.

Appendix E: Constraint algebra and physical degrees of freedom

The combined nonlinear analysis uses a 42-dimensional phase space and 24 constraints. Six constraints are first class and eighteen are second class, giving

$$N_{\text{dof}} = \frac{42 - 2N_{\text{FC}} - N_{\text{SC}}}{2} = \frac{42 - 12 - 18}{2} = 6. \quad (\text{E1})$$

The second-class submatrix was verified to have rank 18 on deterministic generic algebraic patches, while the expected first-class null directions were identified independently. The count is consistent with the nonlinear AeST Hamiltonian analysis of Bataki, Skordis, and Zlosnik [19] and with the auxiliary character of the Type-II acceleration sector [12].

Appendix F: Canonical cubic scale and first-order symplectic reduction

The quadratic principal scalar action defines a canonical field normalization proportional to $M_{\text{Pl}}\sqrt{Z/2}$, where

$$Z = 4K_2 \cosh[(\mathcal{Q} - \mathcal{Q}_0)/Z_0]. \quad (\text{F1})$$

The temporal $F_{\mathcal{Q}\mathcal{Q}\mathcal{Q}}$ vertex, the mixed $\Gamma_{\mathcal{Q}}$ vertex, and the $\mathcal{Y}^{3/2}$ contribution from $J(\mathcal{Y})$ were retained in the cubic normalization. Scanning 10^5 background points over $10^{-12} \leq a \leq 1$ gives

$$\Lambda_{\text{sc,min}} = 1.47423693 \times 10^{-6} \text{ eV} \quad (\text{F2})$$

at $a = 6.44196 \times 10^{-3}$, with the mixed interaction setting the minimum.

The remaining first-order scalar is described by a rank-two second-class pair. After solving the primary constraints,

$$\Theta_{\text{red}} = p_\chi d\chi - \Delta_k \Phi du, \quad (\text{F3})$$

with

$$\Delta_k = ak^2 + \frac{3}{2}I_0\mathcal{Q} \quad (\text{F4})$$

in the conservative matter-free case and an additional positive $(3/2)a^3\rho_m$ contribution in the presence of matter. Thus $\Delta_k > 0$ at $k = 0$ for $I_0, \mathcal{Q} > 0$, while $\Delta_k^{-1} \sim (ak^2)^{-1}$ at large k .

Appendix G: CLASS implementation and numerical regression

The frozen perturbation source used by the implementation is identified by the SHA-256

1b76e97333efd654736519bf324976dba
0906192f674f6d05e74f883f416c479

The mixed-current helper quantities are evaluated directly from the exact background solution. In pseudocode,

```
x    = x0/a^3
FY   = 2*K2*Z0*x/Q
QFY  = 2*K2*Z0*x
```

and the direct $(K_Q - \mathcal{Q}F_{\mathcal{Y}})\chi$ contribution is set to its analytic zero after an independent current-identity audit.

Regression requirements included clean builds, matched homogeneous backgrounds under operator removal, finite CMB and matter spectra, source-diff locality, and a nonzero perturbative response to the mixed operator. Radiation-era histories were checked across multiple initial conditions and expected horizon crossings, and transfer outputs at redshifts 0, 1, 10, 100, 1000 were finite over the solver-supported outputs.

Appendix H: Likelihood bookkeeping and chain audit

The seven independent likelihood contributions are summed directly from the stored chain columns. Aggregate columns are excluded to prevent double counting. At the modified-gravity minimum the contributions are

$$\begin{aligned} \chi_{\text{MG}}^2 &= 583.307420 + 22.665290 + 395.833880 \\ &\quad + 8.7693331 + 9.2113156 + 1405.389200 \\ &\quad + 16.090060 = 2441.2664987, \end{aligned} \quad (\text{H1})$$

and at the Λ CDM minimum,

$$\begin{aligned} \chi_{\Lambda\text{CDM}}^2 &= 586.874460 + 22.731346 + 396.109270 \\ &\quad + 8.8026918 + 10.4193730 + 1407.357200 \\ &\quad + 14.638155 = 2446.9324958. \end{aligned} \quad (\text{H2})$$

The chain convergence values are those quoted in Sec. III. Posterior means are evaluated after the configured burn-in removal.

A fresh runtime replay at the stored minimum coordinates gave $\chi^2 = 2441.26096$ for modified gravity and 2447.10292 for Λ CDM. The release therefore preserves both the original chain likelihood columns and the replay audit. The matched comparison in Table I uses the likelihood values evaluated and stored by the converged production chains, identically for both models.

Appendix I: H_0 prior and profile audit

The original sampling prior was

$$66 < H_0 < 70.5 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (\text{I1})$$

Using the posterior means and standard deviations from the converged chains, the distances from the nearest prior edge are 5.99σ for modified gravity and 6.10σ for Λ CDM. The lower boundaries are still farther away, at 8.76σ and 10.39σ respectively.

The fixed-coordinate profile was evaluated at 41 points from $H_0 = 66$ to $76 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The profile minima occur at the production-chain minimum values. At $H_0 = 70.5$, the increases relative to the replay minima are 331.031 and 258.257 in χ^2 for modified gravity and Λ CDM; at $H_0 = 66$, the increases are 1016.881 and 1191.094.

These values provide a direct diagnostic that the sampled posterior region is not accumulated against either H_0 prior edge.

Appendix J: Reproducibility ledger

A complete reproduction preserves the exact model source, likelihood definitions, nuisance treatment, priors, common convergence criterion, and independent-column

likelihood accounting. The release package contains the manuscript source, BibTeX database, machine-readable likelihood summary, figure inputs, chain-analysis scripts, runtime likelihood injection, and SHA-256 manifests. The development repository is

<https://github.com/KarmirisP/aest-nonseparable-cosmology>.

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